

The Impact of Same-dip Double Subduction on Global-scale Plate Velocity Changes: Insights From Spherical Subduction Models

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Key Points:

- Intra-oceanic same-dip double subduction reorganizes plate motion via torque redistribution
- Geometry exerts first-order control on Euler pole shifts, with rheology setting speeds and modestly modulating shifts
- SDDS initiation or termination offers a potential mechanism for abrupt plate reorganizations

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Abstract

Same-dip double subduction (SDDS), where two parallel slabs dip in the same direction, has been linked to episodes of rapid plate motion in both modern and ancient convergent systems. However, its global-scale kinematic consequences remain poorly constrained. Here we use 3-D spherical geodynamic models to investigate the velocity changes and Euler pole shifts associated with inserting a second slab into an existing subduction system. Across a wide range of rheological and geometric parameters, SDDS consistently increases plate velocity behind the new trench, redirects motion toward the added slab, and produces abrupt Euler pole shifts of up to $\tilde{30}^\circ$. Torque analysis shows that these reorganizations arise primarily from geometric redistribution of basal drag, with rheology modulating absolute speeds but exerting little influence on relative acceleration or pole displacement. These results suggest that intra-oceanic SDDS provides a viable mechanism for abrupt tectonic reorganizations—such as the Late Cretaceous acceleration of Africa and the Eocene Pacific plate motion change—and may represent an important contributor to global plate dynamics.

1 Introduction

Subduction is a crucial process in the evolution of Earth, driving plate tectonics and triggering magmatic and volcanic activity, metamorphism, fluid activity, and seismicity (e.g., Forsyth and Uyeda, 1975; Stern, 2002; Turcotte and Schubert, 2014). Double subduction zones, where two nearby slabs coexist and appear to influence each others geodynamic behavior, are currently prevalent on Earth, e.g. in the Mediterranean (Vignaroli et al., 2008; Peral et al., 2022), Taiwan (Lallemand et al., 2001; Lin & Kuo, 2016), the Molucca Sea (Di Leo et al., 2014; Zhang et al., 2017), the Caribbean Sea (van Benthem et al., 2013; E. M. Conrad et al., 2024), and either side of the the Philippine Sea Plate (Faccenna et al., 2018; Gianni et al., 2025). Additionally, paleo-reconstruction evidence suggests double subduction in systems, such as the Africa-Eurasia convergent system in the late Cretaceous (Gürer et al., 2016, 2022) and the Eocene North Pacific evolution (Domeier et al., 2017; Hu et al., 2022).

Double subduction can occur in several geometries—two slabs dipping toward each other, away from each other, in the same direction, or in opposite directions. Same-dip double subduction (SDDS), in which both slabs dip in parallel, is of particular interest because it is often linked to accelerated plate motion. Notable cases include the Late Cretaceous acceleration of Africa–Eurasia (Gürer et al., 2022) and the rapid motion of the Indian plate during the Cretaceous–Paleogene transition (Pusok & Stegman, 2020). The prevailing explanation for these accelerations is the additional slab pull provided by the second subduction zone, which enhances the net driving force on the overriding or subducting plate. Numerical modeling supports this interpretation (A. Holt et al., 2017; Pusok & Stegman, 2019), consistently showing that SDDS can lead to significantly higher plate velocities than single subduction configurations. More targeted studies, such as those modeling the Neo-Tethyan convergence zone, have demonstrated how same-dip double subduction can explain periods of rapid convergence (Jagoutz et al., 2015; Pusok & Stegman, 2020).

These outcomes can be interpreted within a torque framework, in which driving forces such as slab pull are counteracted by resisting forces like basal drag (Forsyth & Uyeda, 1975; C. P. Conrad & Lithgow-Bertelloni, 2002). In this view, a plate’s Euler pole reflects the equilibrium among these competing torques, and double subduction modifies this balance by redistributing slab pull and effective plate area. However, SDDS does not always manifest as faster convergence. In the Izu–Bonin–Mariana system, for example, the dominant expression is trench migration and stress redistribution rather than a significant increase in plate speed (Cizkova2015,Faccenna2018,Holt2018,Gianni2025). This contrast highlights why intra-oceanic SDDS systems, where a new slab forms within the subducting plate, are better suited for investigating how SDDS reorganizes plate motion.

Accordingly, this study focuses on intra-oceanic SDDS in which a second trench develops within an already subducting plate. This configuration is dynamically distinct because it imposes a complex internal force system and torque redistribution on a single rigid plate, potentially altering velocity. Unlike many regional double subduction systems that involve relatively small plates—e.g., the Molucca Sea Plate, which is only 1000 km across (Zhang et al., 2017)—natural examples of intra-oceanic double subduction often involve very large plates, such as the Pacific Plate or the Neo-Tethyan Oceanic Plate. The dynamic behavior of such large plates under the influence of two subduction zones cannot be effectively captured using oversimplified or small-domain models. Yet, most previous modeling efforts have not fully met this challenge. While 2-D numerical models often employ large plate lengths (e.g., 5000 km in Roy et al. (2024)), they inherently lack lateral stress transmission and mantle flow in the third dimension, both of which are critical for simulating realistic plate behavior. On the other hand, 3-D models, while more complete in dimensionality, often simplify the geometry by letting the second trench slice through the entire plate width (A. Holt et al., 2017), which rarely occurs in nature. Moreover, many 3-D setups use plates with widths of only 1000 km (A. Holt et al., 2017; Pusok & Stegman, 2019), far smaller than real oceanic plates that span over 5000 km in both directions.

Some global-scale modeling efforts—such as that of Hu et al. (2022), which explored the Eocene Pacific plate motion shift—have bridged this gap by integrating plate reconstruction with geodynamic simulations. However, these models are often case-specific and not designed to test the broader parameter space of slab geometries and rheologies, as systematically work in Mishin et al. (2008) and A. F. Holt and Becker (2016), respectively. At the global scale, the sphericity of Earth becomes a non-negligible factor. For plates wider than 2400 km, curvature significantly influences the stress distribution and plate response (Chen et al., 2022; Chaillat et al., 2024). Cartesian models introduce artificial effects at side boundaries, such as unnatural mantle return flows, which are absent in spherical Earth geometry. Spherical models, by contrast, better replicate Earth’s physical conditions and allow for more realistic boundary-free force transmission and plate-mantle coupling.

In this work, we use 3-D spherical numerical models to investigate the influence of same-dip intra-oceanic double subduction zones on global-scale plate velocity. We develop simplified yet geodynamically realistic setups to isolate the impact of the new slab on both convergence rate and directional changes. Our focus is on quantifying how plate motion responds to the geometry and extent of a newly forming slab, as well as variations in crustal, lithospheric, and mantle rheology. A series of numerical experiments is performed to systematically evaluate the resulting velocity patterns and test the feasibility of general plate motion changes induced by intra-oceanic double subduction system. This modeling framework is particularly relevant for understanding past episodes of rapid plate reorganization—such as the Africa–Eurasia convergence acceleration in the Late Cretaceous—and can provide guidance for interpreting future observations of plate motion anomalies.

2 Method

2.1 Model setup

We construct 3-D spherical subduction models (whole mantle domains) using the ASPECT finite element code, version 2.6.0-pre (Bangerth et al., 2023; Heister et al., 2017; Kronbichler et al., 2012). Model geometries, including plate and slab structures, are generated with the Geodynamic World Builder (M. R. T. Fraters et al., 2019; M. Fraters, 2023). Our goal is to characterize plate velocity changes associated with the insertion of a second subduction zone into an existing single-subduction system (A. Holt et al., 2017; Gianni et al., 2025), rather than simulating the detailed initiation process or long-term thermal evolution. A time-dependent approach could capture these processes, but would require significantly more complex rheology and boundary conditions (Mishin et al., 2008; Pusok & Stegman, 2019). Instead, we adopt *pseudo-instantaneous* models, run for only 10 kyr, short

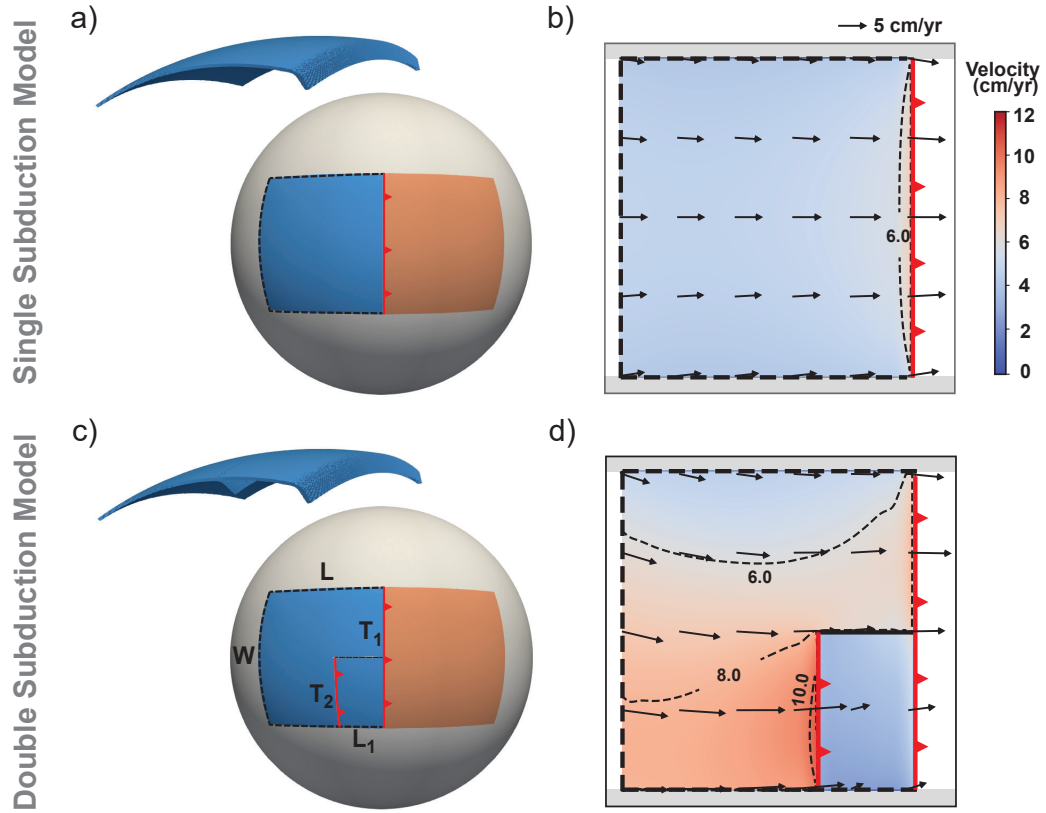


Figure 1. Plate and slab geometries of (a) the reference single subduction model and (c) the reference double subduction model, with (b) and (d) showing their respective plate velocity distributions at 20 km depth. Red lines represent the trench locations and the subducting plates are outlined by black dashed lines. The plate length and width are normalized.

enough that plate and slab geometries undergo minimal evolution from the imposed configuration. This approach allows us to isolate the velocity change associated with transitioning from single to double subduction, under the assumption that plate motion evolves between steady-state configurations. The models are purely compositional: thermal diffusion is neglected, and viscosity and density depend only on compositional fields (A. F. Holt, 2022). Five fields are included: crust and lithosphere of the overriding plate, crust and lithosphere of the subducting plate, lithosphere of the side plate, and weak zone (A. F. Holt et al., 2018). No external forces or prescribed velocities are applied (i.e., the models are fully “dynamic”), and all boundaries are mechanically free slip.

In our reference single slab model (Figure 1a), both the square subducting and overriding plates have a size of 45 degrees ($\approx 5,000$ km). To reduce lateral shortening effects, two half-width side plates are included, separated from the main plate by 50-km-wide weak zones (Yamato et al., 2009). The slab has a radius of curvature of 280 km at the surface, transitioning at 140 km depth into a straight segment dipping 60° to 450 km depth (Lallemand et al., 2005; Cruciani et al., 2005). All plates are 100 km thick, including a 15-km-thick weak crustal layer at the top of both the subducting and overriding plates, and along the upper slab surface to 120 km depth (Figure S1). This crustal layer represents the plate boundary shear zone, which facilitates decoupling between plates and produces more natural velocities consistent with observations (Běhounková & Čížková, 2008; Stegman et al., 2010). In the reference double-slab model (Figure 1c), an intra-oceanic slab is inserted with the same geometry as the main slab, and decoupled from the main plate by a trench-perpendicular, 25 km-wide weak zone cutting through the lithospheric plates and extending between the two trenches. The second trench is located one-third of the plate length away (1700 km) from the main trench, with a trench length of 2500 km (half the main trench length).

Both the lithosphere and crust are assigned constant densities of 3375 kg/m^3 , which is 75 kg/m^3 denser than the sub-lithosphere mantle, that is to provide the negative buoyancy. The lithospheric and crustal layers are isoviscous, and in the underlying sub-lithospheric mantle, we employ a composite rheology combining Newtonian and power-law behavior (Zhong et al., 1998), in order to mimic the first-order effects of diffusion and dislocation creep, respectively. Lithospheric plates are set to be 500 times more viscous ($1 \times 10^{23} \text{ Pas}$) than the reference mantle viscosity ($2 \times 10^{20} \text{ Pas}$), consistent with slab-mantle viscosity contrasts inferred from geoid constraints, slab shapes and trench motions, and plate bending considerations (Billen & Hirth, 2007; Funiciello et al., 2008; W. Schellart, 2009; Ribe, 2010). The crustal layer is assigned a viscosity of $1 \times 10^{20} \text{ Pas}$. Considering the deep plate interface, for example $500\text{--}600^\circ\text{C}$, and strain rate of 10^{-12} s^{-1} , this viscosity is consistent with the recorded blueschists and resides between stronger meta-basalts and weaker metasediments and serpentinites (Behr & Becker, 2018; Tokle et al., 2023), which is also aligned with numerical results (Abila et al., 2024).

Power-law flow is considered due to its large effect on plate and trench motions (Čížková & Bina, 2013; A. F. Holt & Becker, 2016). The Newtonian viscosity component of the mantle, η_N , is set to $4 \times 10^{20} \text{ Pas}$. The power-law component, η_P , is calculated by reducing η_N within high strain rate regions following:

$$\eta_P = \eta_N (\dot{\epsilon} / \dot{\epsilon}_T)^{\frac{1-n}{n}} \quad (1)$$

where $\dot{\epsilon}$ is the square root of the second invariant of the strain rate tensor, n is the power-law exponent ($= 3.5$), and $\dot{\epsilon}_T$ is the transition strain rate, where the strain rate for Newtonian and power-law components are equal, controlling the proportion of the domain within the power-law regime. We assume transition strain rates as $5 \times 10^{-15} \text{ s}^{-1}$ (A. F. Holt & Becker, 2016), then get $\eta_P = \eta_N = 4 \times 10^{20} \text{ Pas}$ at this value. The effective model viscosity is then calculated as:

$$\eta_{eff} = \left(\frac{1}{\eta_N} + \frac{1}{\eta_P} \right)^{-1} \quad (2)$$

which produces the reference mantle viscosity of $\eta_{eff} = 2 \times 10^{20}$ Pas, which is the equivalent value as in Faccenna et al. (2017). Below the 660-km discontinuity, viscosity increases by a factor of 50 relative to the reference upper mantle value (Moresi & Gurnis, 1996).

Given uncertainties in reference viscosities, we test additional values. Relative to reference Crust/Mantle viscosity contrast ($\eta_C/\eta_M = 0.5$), we test a lower value of 0.1 and a higher value of 2. For the Lithosphere/Mantle viscosity contrast ($\eta_C/\eta_M = 500$), we run a lower value of 100 and a higher value of 2000. Relative to reference power-law transition strain rate ($\dot{\epsilon} = 5 \times 10^{-15} \text{ s}^{-1}$), we test low value of 1×10^{-15} and high value of 1×10^{-14} . This verify whether the first-order velocity patterns depend on crustal viscosity, lithosphere viscosity, and mantle viscosity/transition strain rate.

In addition, we vary plate geometric parameters known to affect kinematics, including slab width (W. P. Schellart et al., 2007; Strak & Schellart, 2016), plate length (Xue et al., 2020), trench separation/distance between trenches (Faccenna et al., 2018), and trench obliquity (Balázs et al., 2021), to explore whether the velocity patterns are unchanged. Relative to the reference plate length ($L = 45^\circ/5000 \text{ km}$), we test large value of 54° and small value of 36° , keeping the subducting plate square. For the plate width ($W = 45^\circ/5000 \text{ km}$), we similarly run a large value of 54° and small value of 36° , with the constant plate width. Regarding the trench ratio between the new and old trench length ($T_2/T_1 = 1/2$), we examined large ratio of $2/3$ and small ratio of $1/3$. In our models, $T_1 = W$, because of our geometric setting which the slab extends continuously along the entire subducting plate. Trench separation is normalized by plate length ($L_1/L = 1/3$ in the reference) and varied to $1/2$ and $1/6$. Then, we explore the obliquity of the new trench relative to the original trenches, with values of 0° in reference models, 11° , 22° , and 31° . Finally, we conduct extended tests: in Model M_1 , obliquity is fixed at 31° while plate length L is increased to 60° ; in Model M_2 , trench separation L_1/L is reduced to $1/6$ compared to $1/3$ in M_1 . Full model parameters are listed in Table S1.

2.2 Numerical Parameters

We employ adaptive mesh refinement (AMR) to efficiently resolve both lithospheric deformation and mantle-scale circulation. Refinement is triggered regions with large gradients in viscosity and/or non-zero compositions. This enables us to highly resolve the lithosphere while capturing mantle-scale flow. The refinement strategy produces a maximum refinement corresponding to $\sim 2.8 \text{ km}$ radially thick finite elements (in the lithosphere and crust) and a minimum refinement corresponding to $\sim 45 \text{ km}$ elements (lower mantle). This approach results in $\sim 4,900,000$ cells, distributed non-uniformly across the domain. All simulations were performed on the Skylake (SKX) nodes of the Texas Advanced Computing Center's Stampede3 supercomputer.

2.3 Analysis Method

A significant velocity difference between two models may not solely reflect double subduction related processes; it can also result from variations in mechanical properties, e.g., lithosphere or crustal strength. For instance, a single subduction model with weaker crust or lithosphere may exhibit faster plate motion than a strong double subduction model. To isolate the impact of the second slab, we therefore analyze velocity differences between equivalent double and single subduction (i.e., always with the same mechanical properties), at 20 km depth beneath the highly deformed crustal layer. Specifically, we assess both the ratio of plate velocity magnitudes (e.g., shading in Figure 2a) and the velocity vector (directional) differences (vectors in Figure 2a). Instead of plotting raw differences, we normalize by the corresponding velocity in the single-subduction model. This removes the influence of absolute velocity variations across rheological settings and highlights the relative impact of adding a second trench.

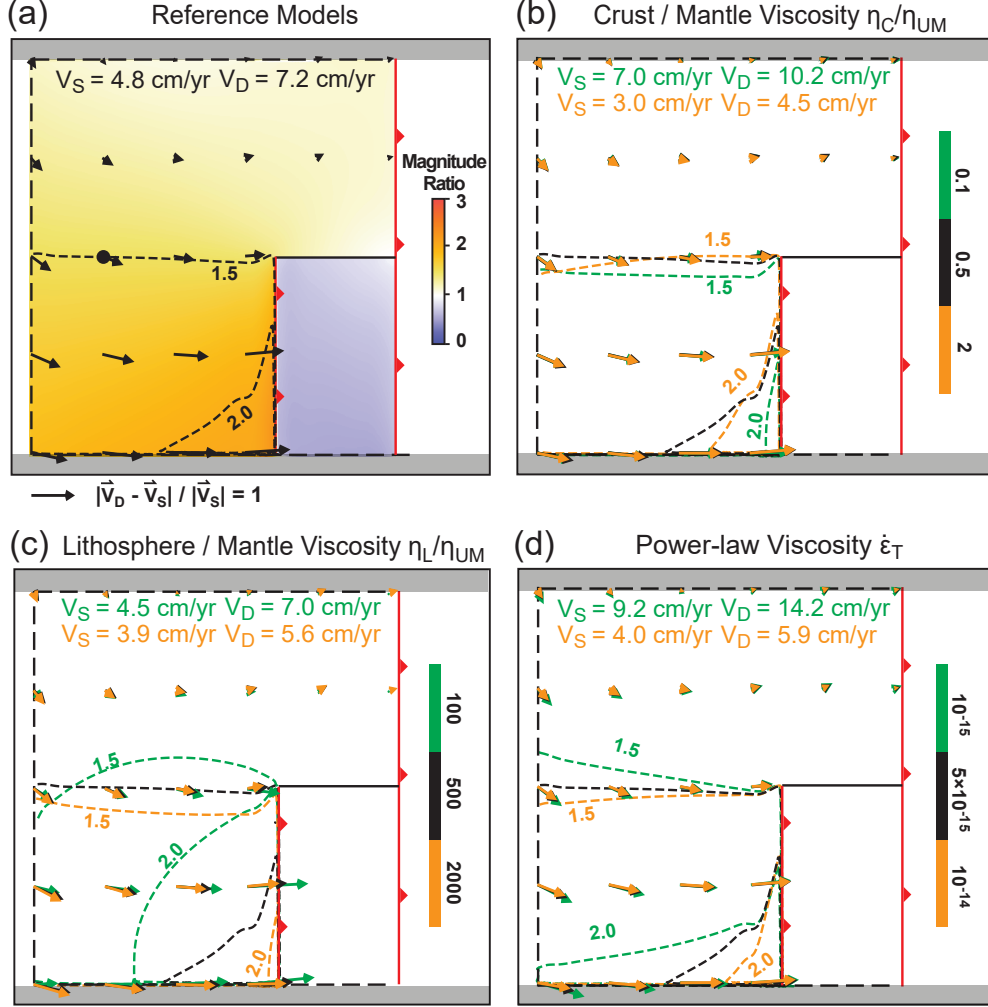


Figure 2. (a) Velocity magnitude ratio (double plate velocity/single plate velocity) for the reference models, with arrows showing velocity difference vectors normalized by local single subduction velocities, $(V_D - V_S)/|V_S|$. V_S and V_D indicate absolute velocities in the single and double subduction models at the location of the black point. Black contours are the magnitude ratio of 1.5 and 2 in reference models. The other panels display velocity difference vectors and contours of the magnitude ratio for pairs of models with variable (b) crust / mantle viscosity ratio, (c) lithosphere / mantle viscosity ratio, and (d) transition strain rate.

To enable consistent comparison of velocity patterns across models with different geometry/sizes, we normalized the subducting plate to a common reference length. In this framework, corresponding points are scaled to the same relative position along the plate, although the absolute distances differ. Velocity vectors are plotted with their original component values and are not artificially stretched by the normalization.

We determine the Euler pole of the subducting plate, which represents the point on the Earth's surface about which the subducting plate undergoes rotation, providing a simplified description of its overall motion. To identify the Euler pole position, we apply a least-squares optimization method that minimizes the misfit between the velocity field predicted from Euler pole rotation and the velocities obtained from the numerical model in the shaded region (e.g., Figure 3a; Liu and King, 2022). After computing the Euler poles for pairs of single and double subduction models, we quantify the shift by calculating the great-circle distance between the corresponding Euler poles (e.g., Figures 3c and 4c). By comparing the computed Euler pole positions with published plate reconstructions (Seton et al., 2012; Butterworth et al., 2014), we can better constrain the reliability of rigid plate motion models and provide potential explanations for noticeable pole position change.

We further analyze the torque balance to interpret Euler pole shifts. Slab pull torque is computed by integrating the force along the subduction zone. (Faccenna et al., 2012). Because the original slab is identical in both single- and double-subduction cases, torque differences arise solely from the added slab. In a steady state, slab pull torque is balanced by basal drag torque, which depends on both velocity and plate area (Iaffaldano & Bunge, 2015). Since the total slab pull force and torque are fixed, accommodating the asymmetric plate geometry caused by the added trench requires a non-uniform velocity distribution, e.g., faster velocities behind the new trench where the plate area is smaller and must accommodate the same amount of torque.

However, such spatial variation is not captured by analytical torque inversion, so we instead use a numerical method to identify the Euler pole that produces a velocity field with basal drag torque balancing the slab pull torque (Figure 7a). In calculating basal drag torque, we exclude the region between the two trenches (non-shaded in Figure 7a), since it forms a separate plate in double-subduction cases. We also introduce a slab pull force transmission factor k , to account for the partial transfer of front slab pull force through the subduction zone, where deformation is complex and the plate may be strongly weakened (Jian et al., 2023). This factor is calibrated by comparing velocity fields and Euler poles from torque calculations with those obtained in the numerical models.

3 Results

3.1 Reference models

We first compare velocity fields between the reference single and double subduction models (Figures 1b and 1d). In the single case, plate velocity reaches ~ 7 cm/yr near the trench and averages ~ 5 cm/yr in the far field. In the double-subduction case, velocities rise to ~ 8 cm/yr near the main trench and ~ 10.5 cm/yr behind the newly formed second trench. Far-field velocities also increase southward, from ~ 5 cm/yr in the single case to ~ 7 cm/yr in the double case. Between the two trenches, velocity decreases relative to the single model, reflecting the formation of a more independent plate bounded by the weak zone.

The velocity ratio (double/single) exceeds 1 across the entire main subducting plate (Figure 2a). The ratio peaks at ~ 2.3 directly behind the new trench and decreases northward and toward the far field. Normalized velocity-difference vectors rotate slightly clockwise, indicating redirection of motion toward the newly inserted slab, with the strongest influence near the southern plate margin. For example, at a representative far-field location (black dot in Figure 2a), velocity increases from 4.8 cm/yr in the single model to 7.2 cm/yr in the double model, giving a ratio of 1.5.

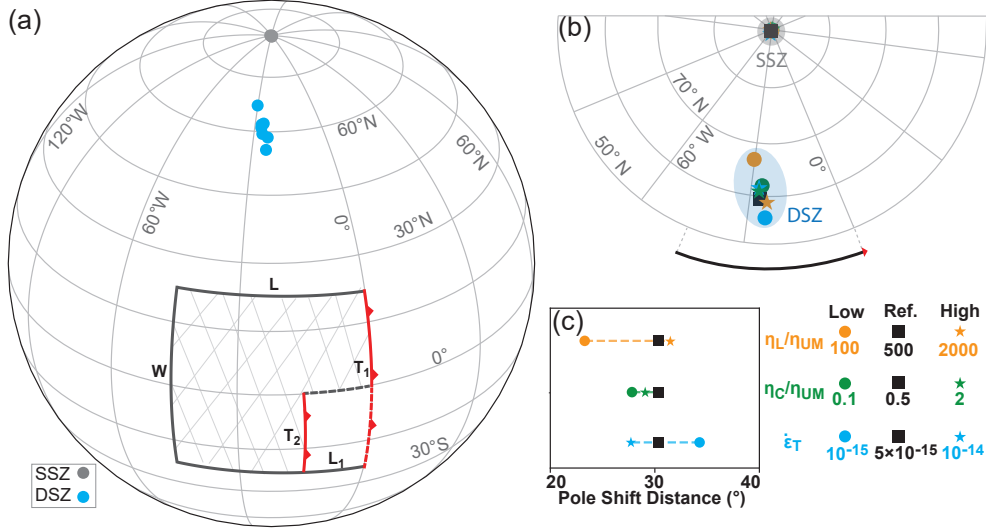


Figure 3. (a) Calculated best-fitting Euler pole locations for single-subduction models (gray) and double-subduction models (blue) with rheological variables. Zoomed-in views of the single-to-double subduction Euler Poles are shown in (b), for pairs of models with the parameters, and the respective Euler Pole great circle shifts are shown in (c). Variables are distinguished by color, and values are represented by different shapes. In panel (b), the black curves with the red arrowhead indicate the range of the subducting plate.

Although our models allow for internal deformation, the velocity fields of the main subducting plate remain close to rigid rotation, as confirmed by a rigidity test (Figure Sx). This justifies the use of Euler poles to characterize plate motion. Best-fitting Euler poles reflect these velocity differences. In the single-subduction case, symmetry keeps the Euler pole at the North Pole. In the double-subduction case, the Euler pole shifts southeastward by $\sim 31^\circ$ along a great-circle path (Figure 3a), consistent with the stronger acceleration at the southern part of the plate.

3.2 Rheological effects on velocity

Systematic variations in crustal, lithospheric, and mantle viscosities show that rheology primarily controls the absolute speed of plate motion but only modestly alters the relative impact of double subduction.

Crustal viscosity exerts the strongest control on absolute plate velocity. In the weak-crust case, the far-field velocity in the double-subduction model reaches 9.2 cm/yr compared to only 4.5 cm/yr in the strong-crust case (Figure 2b). Despite this large contrast in absolute speeds, the double-to-single velocity ratios remain nearly identical across all cases. This indicates that crustal viscosity sets the overall plate speed but does not significantly affect the relative acceleration caused by adding a second trench.

Lithospheric viscosity also influences absolute plate speeds (Figure 2c). Plates with weaker lithosphere move faster in both single- and double-subduction cases and show stronger acceleration that extends farther into the plate interior, while stronger lithosphere restricts velocity enhancement closer to the trench. However, far-field velocity ratios remain broadly unchanged, confirming that lithospheric strength modifies the magnitude of motion but not the fundamental pattern of velocity redistribution.

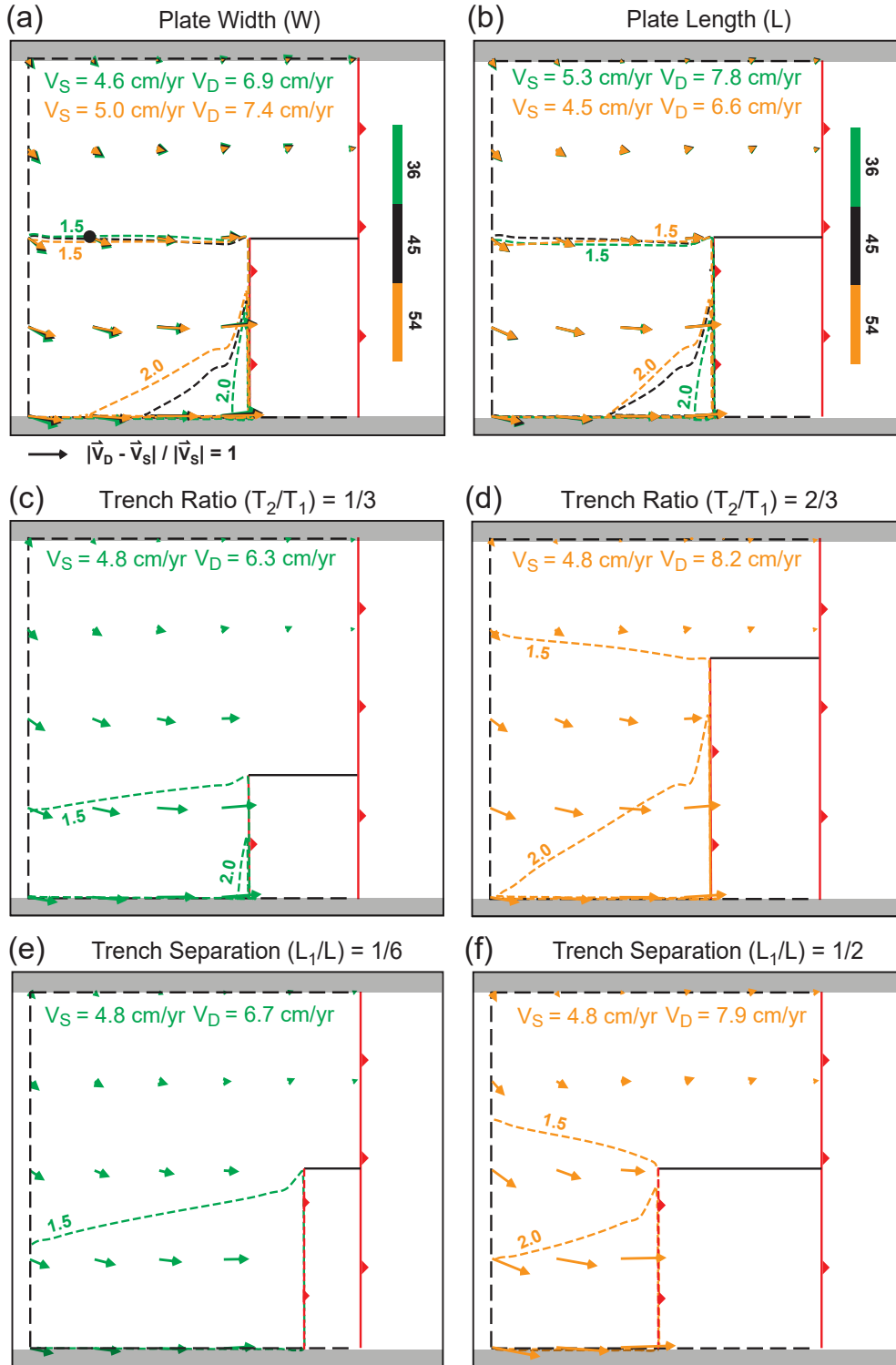


Figure 4. Similar to the previous figure, the panels display velocity difference vectors and contours of the magnitude ratio for pairs of models with variable (a) plate width and (b) plate length. Panels (c) and (d) show low and high values of trench ratio, respectively, while panels (e) and (f) represent low and high values of trench separation.

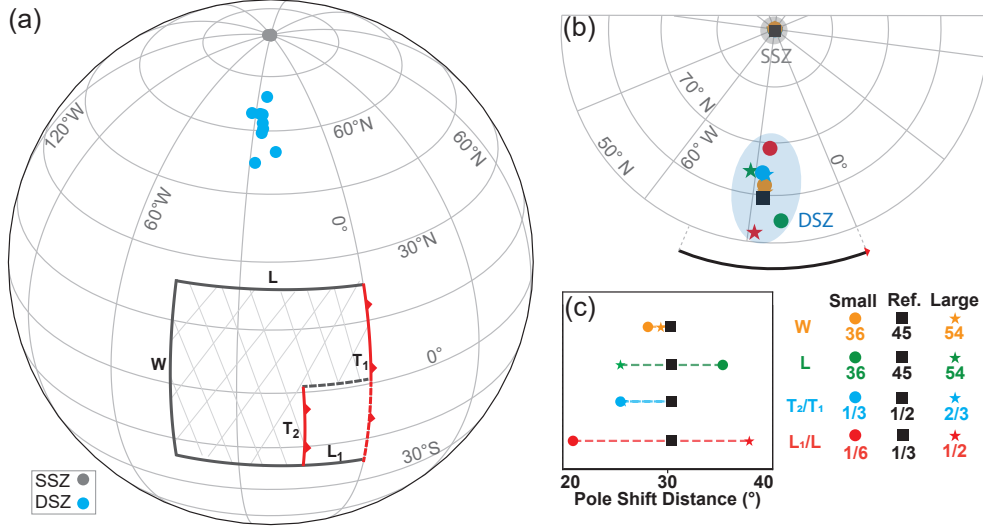


Figure 5. Similar to the previous figure, best-fitting Euler pole positions with plate geometric variables (a), the corresponding zoomed-in view (b), and the computed Euler pole shifts (c).

Mantle rheology controlled by the transition strain rate, similarly affects the velocity scale (Figure 2d). Lower transition strain rates, corresponding to weaker mantle, produce much faster plates in both single and double cases and enhance acceleration at the lateral edges of the plate. This edge effect arises from weak-zone rheology; when a constant-viscosity weak zone is imposed, the boundary acceleration largely disappears (Figure S2). Thus, mantle rheology regulates the efficiency of overall plate motion but leaves the geometry of acceleration patterns intact.

Euler pole positions are less sensitive to rheology than velocities but do show systematic trends (Figure 3). All single-subduction models retain poles at the North Pole, consistent with system symmetry. In double-subduction models, however, Euler poles shift southeastward by 21°–38°. Lithospheric viscosity exerts a measurable influence: the pole shifts by 23° in the weak-lithosphere case, ~30° in the reference, and 31° in the strong-lithosphere case. Mantle viscosity, tuned via the transition strain rate, shows a similar pattern, with the Euler pole shifting 34° in the weak-mantle case (low transition strain rate) and 28° in the strong-mantle case (high transition strain rate). By contrast, crustal viscosity exerts minimal influence on pole location. Taken together, these results show that rheology governs the absolute velocity scale and modulates Euler pole displacement to a limited degree.

3.3 Geometric effect on velocity

In contrast to rheology, geometric parameters exert strong and systematic control on both velocity patterns and Euler pole displacements. Plate width has minimal effect (Figure 4a), but plate length has a pronounced influence: longer plates move more slowly due to greater basal drag, and shorter plates exhibit smaller Euler pole shifts, with the displacement decreasing from 35° in the 54° case to 26° in the 36° case (Figures 4b and 5c). Trench length ratio (T_2/T_1) further controls asymmetry between the two slabs. Larger ratios (2/3) extend the region of enhanced velocity behind the second trench (Figures 4c–d), while smaller ratios (1/3) reduce acceleration. The largest Euler pole shifts occur at the most asymmetric case ($T_2/T_1 = 1/2$; Figure 5c), consistent with torque redistribution driven by uneven trench lengths.

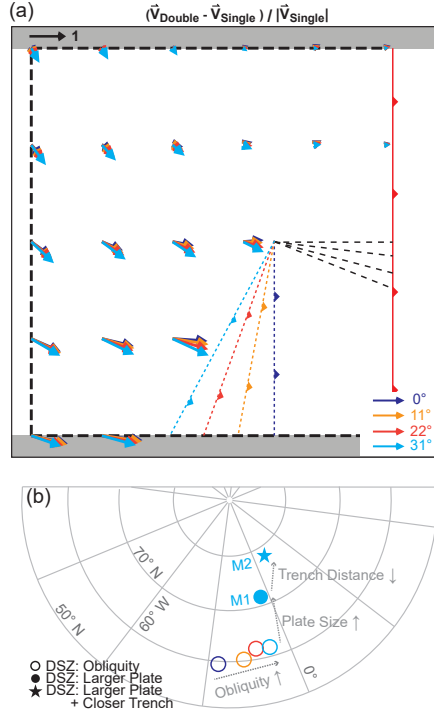


Figure 6. (a) Normalized velocity difference vectors for double subduction models with varying obliquity of the 2nd trench (dashed lines) shown in different colors. The dashed black lines are the corresponding weak zones that are perpendicular to the trenches. (b) Zoomed-in views of Euler poles in models with increasing obliquity (left to right), further increased plate size (M1), and reduced trench separation (M2).

Trench separation exerts the strongest geometric control. With greater separation ($L_1/L = 1/2$), the chosen far-field point lies closer to the second trench, producing broader acceleration and Euler pole displacements up to 38° . By contrast, with smaller separation ($L_1/L = 1/6$), acceleration weakens such that no region reaches a velocity ratio above 2 (Figure 4f), and the Euler pole shift decreases to $\sim 21^\circ$ (Figure 5c). Trench obliquity further alters velocity patterns and pole positions. Increasing obliquity tilts velocity vectors progressively southward by 7° – 14° (Figure 6a) and shifts Euler poles eastward (Figure 6b). Extended tests illustrate combined effects: in Model M_1 , where plate size is increased under 31° obliquity, the Euler pole lies closer to the single-subduction location, while in Model M_2 , where trench separation is reduced relative to M_1 , the Euler pole displacement is smaller still (Figure 6b). Overall, geometric parameters — especially trench separation and trench-length asymmetry — dominate the redistribution of plate velocities and the reorganization of Euler pole positions, with plate width exerting little influence and obliquity producing secondary but systematic eastward deflections.

3.4 Torque balance

Torque computations provide a mechanical explanation for the observed reorganizations (Figures 7–8). In the single-subduction case, slab pull and basal drag torques balance at the North Pole. Adding a second trench shifts the integrated slab pull torque southward by $\sim 2^\circ$ (Figure 7a), requiring Euler pole displacement to restore equilibrium. Decomposition shows that reduced effective plate area contributes $\sim 21^\circ$ of Euler pole shift, while the added

slab pull from the second trench accounts for $\sim 12^\circ$. Together, these contributions sum to the $\sim 31^\circ$ displacement observed in the reference model (Figure 7b). Across parameter tests, the slab-pull component remains relatively constant, while variations in effective plate area drive most of the variability in total displacement.

Geometric variations redistribute torque in predictable ways. Increasing plate length shifts the torque southwestward, reducing the Euler pole displacement (Figure 7c). Decreasing trench separation redistributes torque primarily in longitude, producing smaller pole shifts (Figure 7d). Increasing obliquity rotates the torque vector southeastward, though in these cases analytical torque balance diverges more strongly from numerical Euler pole positions, reflecting reduced efficiency of force transfer across oblique interfaces (Figure 7e). Force-transmission efficiency is quantified by the factor k . A value of $k = 0.3$ reproduces most velocity and Euler pole results, consistent with effective slab-pull transfer across relatively stable interfaces, whereas smaller values are required for oblique geometries where force dissipation is greater (Figure 8). These torque results quantitatively link geometry-driven velocity increases and Euler pole displacements to underlying force balances, reinforcing that same-dip double subduction reorganizes plate motion primarily through geometric redistribution of basal drag.

4 Discussion

4.1 Mechanism of velocity changes

Our main objective is to test how intra-oceanic same-dip double subduction alters plate velocities and Euler pole positions across a wide range of rheological and geometric conditions, and to connect these changes to plate reorganizations in Earth history. Slab pull has long been recognized as the dominant driver of plate motions, with Euler pole positions reflecting the balance between driving and resisting torques (Forsyth & Uyeda, 1975; Lithgow-Bertelloni & Richards, 1998; C. P. Conrad & Lithgow-Bertelloni, 2004). Global torque budget and mantle flow studies further show that this balance depends strongly on subduction zone geometry and mantle rheology (C. P. Conrad & Lithgow-Bertelloni, 2002; Becker & O’Connell, 2001), while more recent work emphasizes that temporal changes in slab distribution can reorganize torques and alter plate kinematics (Austermann et al., 2011). To isolate this effect, we kept slab properties fixed (dip, density contrast, and thickness) across all experiments, so that any variation in torque arises from geometric redistribution of basal drag and the relative contribution of the additional slab. This design directly links plate geometry to the velocity changes and Euler pole shifts observed in our experiments.

Figure 8 demonstrates how torque redistribution explains the Euler pole shifts in our models. Panels 8a–c show systematic correlations between Euler pole displacement and geometric parameters. Increasing plate length amplifies the shift (Figure 8a), smaller trench separations suppress it (Figure 8b), and greater obliquity deflects poles eastward and southward (Figure 8c). These results confirm that geometry exerts the first-order control on Euler pole positions, consistent with studies highlighting the role of slab width, trench length, and subduction partitioning in controlling plate kinematics (W. P. Schellart et al., 2007; Strak & Schellart, 2016; Faccenna et al., 2018).

Panels 8d–f decompose the torque balance into contributions from reduced plate area and added slab pull. Both components are required to reproduce the observed shifts: reduced plate area accounts for $\sim 21^\circ$, while added slab pull contributes $\sim 12^\circ$, together matching the $\sim 31^\circ$ displacement observed in the reference model (Figure 7b). Across variable tests, the slab-pull contribution remains relatively steady, while changes in plate area generate the largest differences in shift magnitude. In all cases, the total Euler pole displacement closely approximates the sum of the two contributions, underscoring that geometric redistribution of basal drag is the dominant factor controlling variability. For plate length and trench separation, a constant transmission factor of $k = 0.3$ reproduces the numerical results, but

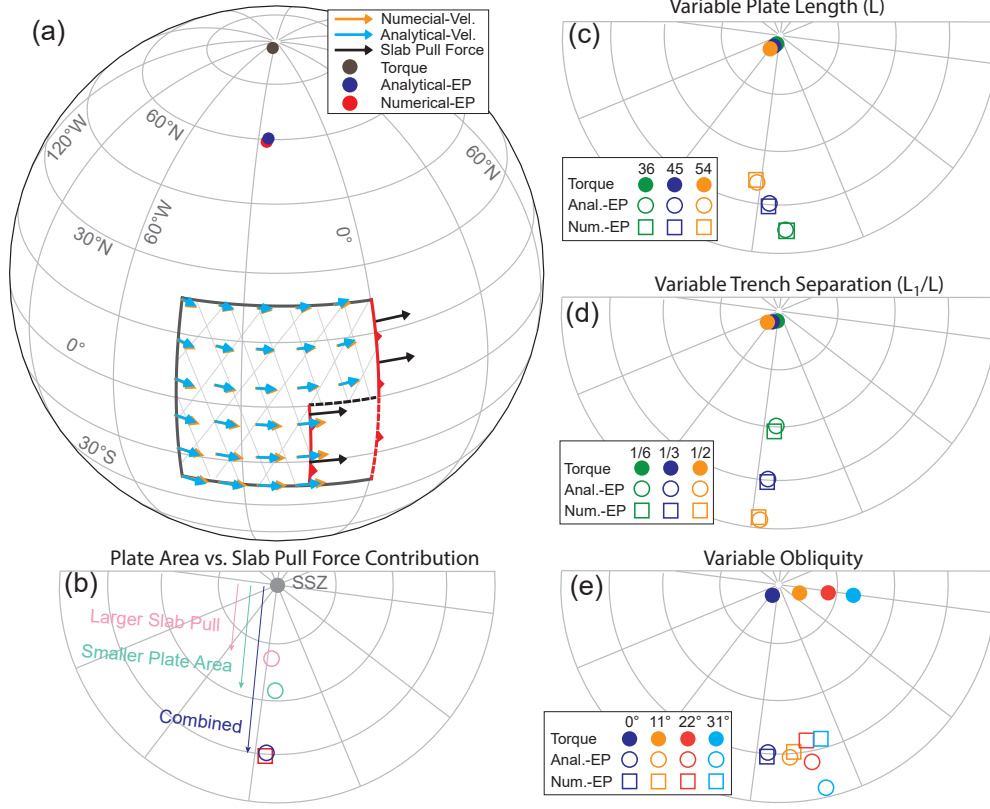


Figure 7. Comparison of torque and Euler pole positions in the reference models (a). Euler pole shift contributions from additional slab pull force and reduced plate area (b). Torques and Euler pole shifts with varying plate length L (c), varying L_1/L (d), and varying obliquity of the new trench (e). The torque positions represent both slab pull integrated torque and basal drag torque. Analytical-EP indicates the Euler pole position producing the velocity field satisfying the balance between integrated drag basal torque and slab pull torque. Numerical-EP is the position of the best-fitting Euler pole directly computed from the model velocity. Numerical-Velocity and Analytical-Velocity are the velocities output from numerical models and the torque balance, respectively.

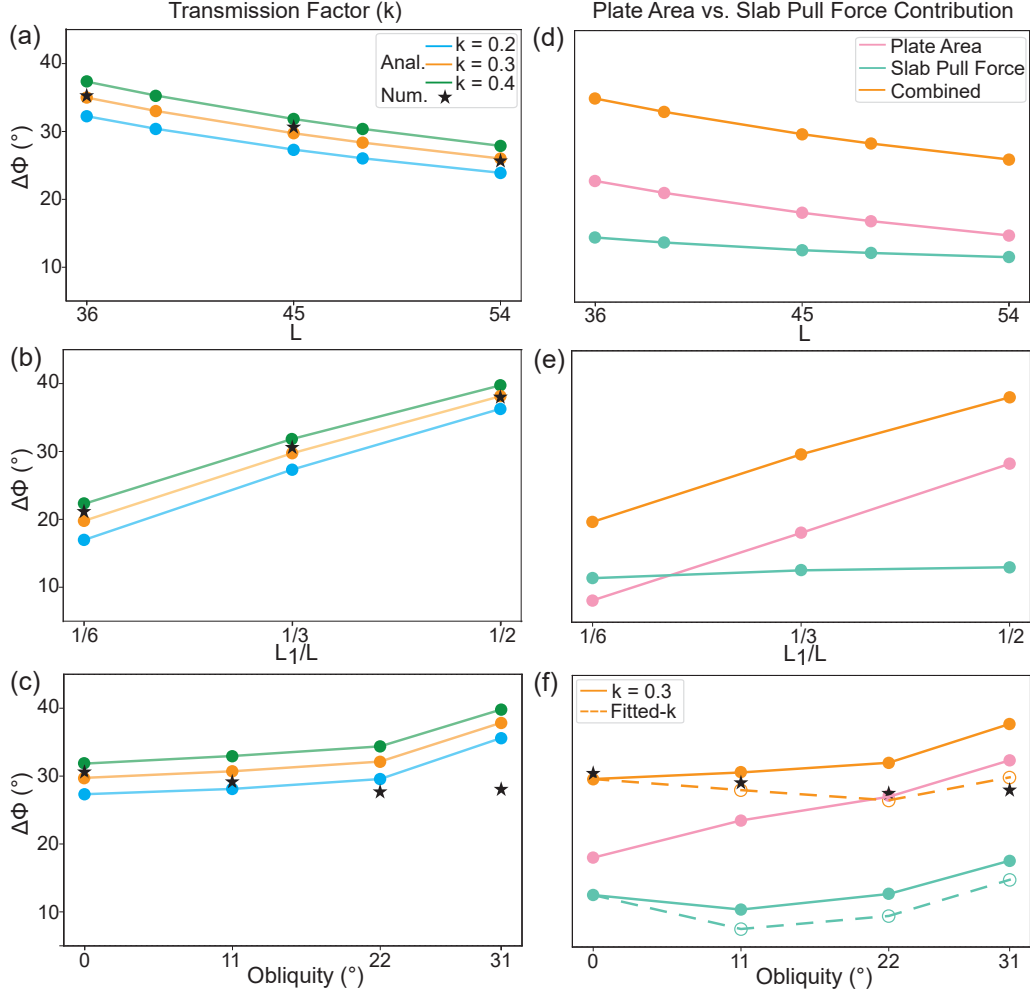


Figure 8. Euler pole shifts as a function of (a, d) plate length (L); (b, e) trench separation (L_1/L); (c, f) obliquity. Panels (a-c) show results under varying force transmission factor $k = 0.2$ (blue), 0.3 (orange), and 0.4 (green). Panels (d-f) show results at a constant $k = 0.3$, with curves representing the contribution from reduced plate area (pink), additional slab pull force (cyan), and the combined effect (orange). Stars are the corresponding numerical Euler pole shift values. Dashed curves in (f) show different k values that align the total Euler pole shifts with numerical results.

smaller values are required for obliquity tests. This indicates that slab pull is transmitted efficiently across relatively stable interfaces but less so when the intervening interface is oblique or extended, where more force is dissipated locally. Previous 3-D models (A. Holt et al., 2017) and recent regional studies of the Philippine Sea system (Jian et al., 2023) similarly demonstrate that slab-pull transmission is incomplete and highly sensitive to geometry and rheology. Our results extend this framework by quantifying how obliquity within same-dip double subduction further reduces transmission efficiency.

While torque redistribution controls the relative reorganization of motion between single and double subduction states, the absolute plate speeds in our models are highly sensitive to both rheology and geometric configuration, with viscosity contrasts setting the overall velocity scale and parameters such as trench separation modulating the degree of basal drag resistance. Stronger lithospheres increase bending resistance and suppress velocities,

while weaker lithospheres promote faster subduction (Buffett, 2006). Similarly, a stronger lithosphere–asthenosphere viscosity contrast enhances decoupling and speeds up motion, whereas higher asthenospheric viscosity reduces plate velocity (Funiciello et al., 2008; A. Holt et al., 2017). In our tests, crustal viscosity primarily controls absolute plate speeds (Behr & Becker, 2018), while lowering mantle viscosity (via reduced transition strain rate) enhances acceleration in the double-subduction–dominated region (Doglioni et al., 2011).

These effects are well captured by a simplified 2-D force-balance analysis (Text S1). For single subduction, slab pull resisted by bending, basal drag, and interface forces yields a predicted velocity of ~ 4.8 cm/yr, consistent with the reference numerical model. For double subduction, if 30% of the front-slab pull is transmitted across the plate ($k = 0.3$), the predicted velocity is ~ 8.5 cm/yr, giving a double-to-single ratio of ~ 1.8 . This matches our numerical outcomes and falls within the range reported in laboratory and numerical experiments (Funiciello et al., 2008; Čížková & Bina, 2013; A. Holt et al., 2017; Pusok & Stegman, 2019), reinforcing that velocity enhancement is a robust consequence of torque redistribution. Variations in trench separation also affect the absolute speeds: the computed double-to-single velocity ratio increases to ~ 2 for wider separation ($L = 1/2$) and decreases to ~ 1.6 for narrower separation ($L = 1/6$). This trend, where greater separation produces faster double-subduction velocities, is consistent with the scaling expected from basal-drag resistance and agrees with both our model results and previous findings (Faccenna et al., 2018).

In summary, the mechanism underlying our results is that geometric redistribution of basal drag torque, under relatively constant slab pull, reorganizes plate velocities and Euler poles. These insights set the stage for interpreting geological examples of abrupt plate motion changes.

4.2 Implications for plate reorganizations

Understanding these dynamics is crucial because they provide a physical mechanism for major plate reorganizations observed in Earth history. Several events display signatures reproduced by our models: acceleration with slab insertion, deceleration with slab loss, and associated Euler pole shifts. For example, in the Late Cretaceous, Africa accelerated from ~ 20 – 30 to 45 – 80 mm/yr following intra-oceanic subduction initiation in Neo-Tethys, and later slowed during obduction and slab termination (Jagoutz et al., 2015; Gürer et al., 2016; van Hinsbergen et al., 2021; Gürer et al., 2022). In the Pacific, the Hawaii–Emperor bend coincided with initiation of the Izu–Bonin–Mariana system and possible termination of the Kronotsky slab (Whittaker et al., 2007; Domeier et al., 2017; Hu et al., 2022). Comparable reorganizations have also been recognized in global reconstructions (Seton et al., 2012; Müller et al., 2016).

Our models reproduce the qualitative fingerprints of these events, though Euler pole displacements in models exceed typical reconstructed values (e.g., 3°). This mismatch emphasizes that natural settings involve additional complexities—such as larger plates, greater obliquity, and smaller trench separations—that can reduce the magnitude of reorganization. Nevertheless, the modeled trends highlight same-dip double subduction as a viable mechanism for abrupt tectonic reorganizations. Crucially, by holding slab properties constant across all experiments, we isolate the role of plate geometry and avoid the slab-pull force variations inherent to time-dependent models. In contrast to previous studies that coupled torque redistribution with evolving slab dynamics, our results demonstrate that geometry alone can reorganize torque balance, accelerate plate motion, and drive abrupt Euler pole shifts. This highlights a fundamental mechanism for plate reorganizations and explains why same-dip double subduction is particularly effective in producing them.

Beyond these case studies, the broader lesson is that reorganizations in global plate motion can emerge purely from changes in torque balance. This view complements reconstructions attributing reorganizations in the Pacific and Africa to slab-geometry changes

(Whittaker et al., 2007; Butterworth et al., 2014) and supports the perspective of a coupled plate–mantle system where self-organized torque evolution governs plate dynamics (C. P. Conrad & Lithgow-Bertelloni, 2002; Cramer et al., 2019; Coltice et al., 2019).

Our analysis is restricted to intra-oceanic, same-dip double subduction, chosen because this configuration is most directly linked to changes in plate velocity and Euler pole position. Other double-subduction geometries, such as divergent or convergent systems, are expected to produce different velocity fields and torque responses, but remain insufficiently explored. Extending such tests, along with incorporating additional complexities such as thermal evolution, pressure-dependent viscosity, and time-dependent slab growth (Tackley, 2000; Billen, 2008) will be essential for evaluating the persistence of these mechanisms under more Earth-like conditions. Even in their simplified form, however, our models reproduce the essential dynamical fingerprints of natural reorganizations, including abrupt Euler pole jumps, acceleration behind new trenches, and velocity reorientation toward them.

5 Conclusions

Our 3-D spherical geodynamic models show that intra-oceanic same-dip double subduction (SDDS) reorganizes global-scale plate kinematics by accelerating motion behind the new trench, redirecting velocities toward the added slab, and driving abrupt Euler pole shifts. These effects are robust across a range of rheological parameters, indicating that geometry exerts the first-order control while rheology primarily sets the absolute velocity scale. Torque analysis demonstrates that reduced effective plate area and additional slab pull together account for the observed Euler pole displacements, linking the reorganizations to geometric redistribution of basal drag.

These findings suggest that the initiation or termination of intra-oceanic SDDS may provide a physical mechanism for abrupt changes in plate motion. Natural examples, such as the Late Cretaceous acceleration of Africa and the Eocene Pacific reorganization, exhibit signatures consistent with our modeled outcomes, though additional complexities in plate size, obliquity, and slab dynamics likely modulate the response. We therefore propose intra-oceanic SDDS as a potential contributor to past tectonic reorganizations and as a framework for interpreting future anomalies in global plate kinematics.

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: [https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors](https://www.agu.org/Publish-with-AGU/Publish/Author-Resources/Data-for-Authors)

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